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Information handling system keyboard support with rail guide structure assembly to a housing rail

Abstract

A portable information handling system is assembled and disassembled by a keystone assembly that couples at an upper surface of the portable housing to overlap a keyboard assembly that covers processing components disposed in the portable housing. The keyboard assembly includes a housing cover having guide structures with stops disposed on opposing sides and aligned to engage with rails of the housing to slide to a cover position over the housing defined by the position of the stops. The keystone assembly completes the housing cover enclosure of the housing interior by coupling in a gap between the housing cover and housing at one side and overlapping the keyboard assembly to retain the keyboard assembly in position.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates in general to the field of portable information handling systems, and more particularly to an information handling system keyboard support with rail guide structure assembly to a housing rail.

Description of the Related Art

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

(3) Portable information handling systems integrate processing components, a display and a power source in a portable housing to support mobile operations. Portable information handling systems allow end users to carry a system between meetings, during travel, and between home and office locations so that an end user has access to processing capabilities while mobile. Tablet configurations typically expose a touchscreen display on a planar housing that both outputs information as visual images and accepts inputs as touches. Convertible configurations typically include multiple separate housing portions that couple to each other so that the system converts between closed and open positions. For example, a main housing portion integrates processing components and a keyboard and rotationally couples with hinges to a lid housing portion that integrates a display. In a clamshell configuration, the lid housing portion rotates approximately ninety degrees to a raised position above the main housing portion so that an end user can type

inputs while viewing the display. After usage, convertible information handling systems rotate the lid housing portion over the main housing portion to protect the keyboard and display, thus reducing the system footprint for improved storage and mobility.

(4) Conventional design goals for portable information handling systems have attempted to compress as much processing capability into as small of a housing footprint as possible. These conventional design goals can result in complex systems where design constraints include substantial processing capability, maximum thermal characteristics, minimal power consumption with maximum battery life, low weight and minimal housing height. Typical designs include specialize housing arrangements assembled with screws that attempt to optimize the use of the entire interior housing space taking into account airflow for cooling, battery dimensions, keyboard height processing component dimensions, display dimensions and other factors. Assembly of the housing and processing components with screws helps to ensure a robust system that can withstand impacts common to portable housing configurations. Generally, conventional designs involve a final assembly that is not intended to be broken down except for the most common repairs, such as replacement of a keyboard, addition of DRAM modules and addition or replacement storage devices. Even these types of interactions typically involve removal of dozens of screws and separation of communication cables that interface the various processing components.

(5) Another difficulty with conventional portable information handling system design is that systems are difficult to breakdown at system end-of-life to support recycling and/or reuse of components. Screws, in particular, tend to make system breakdown difficult and inefficient so that recycling can often become cost prohibitive. For example, a portable information handling system at end of life typically has some components that may have additional reuse life remaining, some components that have recyclable materials and some components that are waste. The value of any reusable and recyclable materials is reduced by the cost of separating those materials from waste materials in a form that allows reuse and recycling. When system breakdown cost exceeds the value of reuse and recycling, a default response is too often that components and materials having value are instead sent to waste. Yet, even in those instances where part of the system components are separated to recycle or reuse, many of the system-specific components will not adapt to new use models. For instance, specialized arrangements of components typically result in cables and connectors of different lengths and types to interface the components within a housing. Often the cable format remains the same across information handling system platforms, yet the cables of different lengths make reuse difficult and the cable connectors couple to circuit boards that are difficult to breakdown and reuse.

SUMMARY OF THE INVENTION

(6) Therefore, a need has arisen for a system and method which assembles information handling system portable platform architecture suitable for efficient system break down that encourages recycling and reuse.

(7) In accordance with the present invention, a system and method are provided which substantially reduce the disadvantages and problems associated with previous methods and systems for assembly and disassembly of a portable information handling system. A portable housing couples components without screws in an architecture adapted for component reuse, refurbishment and recycling using automated methods, such as robotic arms.

(8) More specifically, a portable information handling system housing couples together with a single keystone assembly that releases with a sliding motion adaptable to automated manufacturing techniques. Release of the keystone assembly allows a housing cover to slide relative to a main housing portion and lift off to expose internal components for access by robotic arms. A keyboard assembly includes a membrane having key input sensors, a capacitive touch sensor, speaker signal interface wires and battery enable wires include and managed by a printed circuit board integrated in the membrane. The printed circuit board folds to a bottom side of the housing cover to align with a keyboard connector on the system motherboard so that the keyboard interface with the

motherboard is completed without an internal cable. The motherboard couples to the housing by sliding onto retention nuts and is secured from disassembly by placement of another component to prevent the sliding, such as a cooling fan that couples up against a side of the motherboard. By avoiding cables and screws, the system assembles and disassembles with mechanical manipulation of automated robotic arms so that components are automatically sorted according to available reuse, refurbishment and recycling classifications.

(9) The present invention provides a number of important technical advantages. One example of an important technical advantage is that a portable information handling system assembles and disassembles in a manner consistent with the operation of automated or robotic techniques. Components couple into place and release from an assembly with simplified manipulation, such as sliding of a motherboard or housing cover to fix into place and release from an assembly. Components assemble and breakdown in logical elements associated with reuse, refurbishment and recycling of the components, such as types of material used, life cycle of the component, and availability for refurbishment. For instance, a keyboard breaks down for separate management of a lattice of keys, a membrane with key sensors and a housing cover. The membrane includes wirelines that replace cables used in conventional information handling systems, such as to support communication of speaker signals and battery power. A single keystone assembly completes system assembly and initiates system disassembly without the use of any screws in any of the major defined component elements. Reducing coupling structures, cables and connectors inside a portable information handling system encourages a circular economy that reduces emissions and waste associated with manufacturing, reduces the carbon footprint and enables automated recycling and reuse to achieve improved efficiency and product use for a full life cycle of the components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention may be better understood, and its numerous objects, features and advantages made apparent to those skilled in the art by referencing the accompanying drawings. The use of the same reference number throughout the several figures designates a like or similar element.

(2) FIG. 1 depicts a front perspective view of a portable information handling system configured to promote automated assembly and disassembly to assist with reuse and recycling of information handling system components;

(3) FIGS. 2 and 2A through 2D depict an example of a keystone assembly system for assembly and disassembly of a portable information handling system;

(4) FIG. 3 depicts an upper perspective view of the main housing portion when the housing cover lifts free after removal of the keystone assembly;

(5) FIGS. 4 and 4A through 4C depict a lower rear perspective and cross-sectional view of the housing cover coupling components within the main housing portion by lock members extending down from the housing cover;

(6) FIG. 5 depicts a bottom perspective view of the housing cover configured to couple in a sliding relationship with a main housing portion;

(7) FIGS. 6A and 6B depict a detail view of the housing cover having stops that complete coupling of the housing cover to main housing portion guides;

(8) FIGS. 7, 7A and 7B depict a speaker having speaker signals to play audible sounds communicated through a keyboard membrane and isolated spring biased contact pins;

(9) FIG. 8 depicts an exploded view of a keyboard having a keyboard membrane that includes key touch sensors, a touchpad sensor and a printed circuit board to manage logical tasks associated with keyboard;

- (10) FIG. 9 depicts an upper perspective view of a keyboard membrane having an integrated printed circuit board with a keyboard matrix controller and capacitive touch controller to manage keyboard and touchpad operations;
 - (11) FIGS. 10 and 10A through 10C depict example embodiments of keys assembled as a lattice with a planar sheet between the keys and the membrane to enhance key and keyboard reuse;
 - (12) FIG. 11 depicts an upper perspective view of a keyboard membrane and example embodiment having liquid detection sensors;
 - (13) FIG. 12 depicts a block diagram of liquid detection logic to detect and store a liquid spill event at a portable information handling system;
 - (14) FIG. 13 depicts a block diagram of an intrusion detection circuit to detect and store intrusion events within the system housing;
 - (15) FIGS. 14 and 14A depict an example of disposition of keyboard assembly to motherboard connectors with intrusion detection;
 - (16) FIGS. 15 and 15A through 15C depict an example of a 12 pin keyboard membrane connector interface with integrated intrusion detection;
 - (17) FIG. 16 depicts a circuit block diagram of interfaces between an information handling system motherboard and keyboard printed circuit board;
 - (18) FIG. 17 depicts an upper perspective exploded view of the keyboard assembly and disassembly of a palm rest with electrically reactive adhesive;
 - (19) FIGS. 18 and 18A depict a circuit block diagram of keyboard assembly and disassembly of a palm rest with electrically reactive adhesive;
 - (20) FIGS. 19A through 19C depict a cooling fan coupling with a motherboard in a main housing portion to secure the motherboard in place;
 - (21) FIGS. 20 and 20A through 20E depict an information handling system having a series cooling fan interconnect with adjustable cooling capacity;
 - (22) FIG. 21 depicts a circuit block diagram of an example of communication paths between serially interfaced cooling fans;
 - (23) FIGS. 22, 22A and 22B depict an example embodiment of a cable-free battery connector to communicate power between an information handling system motherboard and battery;
 - (24) FIG. 23 depicts an exploded front perspective view of the cable-free battery connector;
 - (25) FIG. 24 depicts a front perspective view of the battery connector coupled to the main housing portion and having the motherboard coupled in place to electrically interface with the battery;
 - (26) FIGS. 25, 25A and 25B depict an example of an assembly of components with retention nuts 320;
 - (27) FIG. 26 depicts an antenna module integrated in a modular thermal solution;
 - (28) FIGS. 27A, 27B and 27C depict an alternative embodiment for coupling a motherboard to the information handling system main housing portion;
 - (29) FIGS. 28A and 28B depict an example of a flexible slide motion arrangement for coupling a motherboard to the main housing portion; and
 - (30) FIG. 29 depicts a bi-stable flexure plate for coupling a motherboard to a main housing portion.
- DETAILED DESCRIPTION**

(31) A portable information handling system design shifts towards a reduced carbon footprint with components that enhance repair, refurbishment, material recovery and harvesting for reuse to accelerate the circular economy, reduce emissions in manufacturing and promote recycling and reuse through automation. For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer, a network storage device, or any other suitable device and may vary in size, shape, performance,

functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

(32) Referring now to FIG. 1, a front perspective view depicts a portable information handling system **10** configured to promote automated assembly and disassembly to assist with reuse and recycling of information handling system components. Portable information handling system **10** is built in a housing **12** having a clamshell configuration with a main housing portion **14** rotationally coupled to a lid housing portion **16** by hinges **18**. In the open position depicted by FIG. 1, lid housing portion **16** is rotated to a viewing position above housing cover **20** that supports a keyboard **22** in a typing position to accept end user presses at keyboard keys. A touchpad **24** is defined in the housing cover **20** to accept end user touches as inputs, such as to control a cursor position or accept an “enter” input. A locking device **26** couples into a security opening of housing **12** to secure the system, such as a Kensington lock and cable arrangement. A display **28** integrates in lid housing portion **16** to present information as visual images viewable to an end user typing at keyboard **22**. Portable information handling system **10** supports portable operating modes with an integrated input device, display and power source assembled in an integrated unit that is disassembleable with an automatic or robotic operation that does not involve screws or other coupling devices needing manual manipulation. The components assemble and breakdown in a manner that promotes reuse and recycling. In the depicted configuration, with security lock **26** coupled in place, the assembled system is protected against internal access. Yet, when security lock **26** is removed and an internal keystone assembly is activated, the entire information handling system breaks down into logical components with minimal time and fully accessible to robotic-type of tools.

(33) Referring now to FIGS. 2 and 2A through 2D, an example of a keystone assembly **30** system is depicted for assembly and disassembly of portable information handling system **10**. FIG. 2 depicts portable information handling system **10** in tablet mode having hinges **18** rotating lid housing portion **16** 180 degrees relative to main housing portion **14** so that display **28** lays flat on a support surface to accept touch inputs. Keyboard **22** and touchpad **24** are supported on the upper side of main housing portion **14** and housing cover **20**. On a side opposite security lock **26** a pair of data cables **34** fit into ports **32**, such as USB Type C cables and ports that support power, video and data communication. In the tablet configuration a keystone assembly **30** couples to main housing portion **14** and overlaps keyboard **22** and cover housing portion **20** to prevent disassembly of portable information handling system **10**. Security lock **26** inserts through a security opening of main housing portion **14** to engage a security lock device **36** coupled at the interior of housing **12** so that a cable **38** secures portable information handling system **10**. For example, security lock **26** locks and unlocks in response to a key actuation with key actuation to a locked position engaging security lock device **36** to prevent access to the security opening in housing **12**. In the locked position, a latch or other coupling arrangement of security lock **26** and security lock device **36** prevents removal of security lock **26** until the lock is moved to an unlock position, such as a Kensington or Noble lock or similar arrangement. When security lock **26** is unlocked to release it from security locking device **36** and removed from the security opening of housing **12**, access is provided to a release of keystone assembly **30** so that housing **12** may be disassembled.

(34) FIG. 2A depicts a perspective view of the inner side of keystone assembly **30**. Keystone assembly **30** has an exterior member **40** sized to fit in a gap between housing cover **20** and the perimeter of main housing portion **14** along a side at which hinges **18** couple. A side wall **54** extends down from opposing lengths of exterior member **40** sized to fit an interior member **42** on

the interior side of exterior member **40** so that interior member **42** slides between side walls **54**. Guide posts **44** extend down from the interior side of exterior member **40** to fit in guides formed in the interior of main housing portion **14**. The engagement of guide posts **44** in guides of main housing portion **14** holds exterior member **40** in a fixed location relative to main housing portion **14**, such as to fill the gap at the rear side of main housing portion **14** perimeter. A latch **46** and a ramp member **48** extend down from interior member **42** into main housing portion **14** to engage with a catch and a ramp coupled to main housing portion **14**. Latch **46** is a resilient material that snaps into place against a catch and has a lip to engage the catch and hold both interior member **42** and, through side walls **54**, exterior member **40** to main housing portion **14**. When interior member **42** slides relative to exterior member **40**, latch **46** moves laterally to release the main housing portion catch and ramp member **48** slides laterally to work against a ramp surface within main housing portion **14** so that exterior member **40** is pushed out of and away from main housing portion **14**.

(35) FIGS. 2B and 2C show detailed perspective views as indicated in FIG. 2A. A release member **56** extends down from interior member **42** at the security opening of main housing portion **14** to accept a press from an external release member inserted through the security opening. When release member **56** is pressed inward, interior member **42** slides relative to exterior member **40** so that latch **46** and ramp member **48** move relative to main housing portion **14**. A spring element **50** formed in interior member **42** biases against at stop **52** of exterior member **40** so that interior member **42** is biased to a locked position with latch **46** position to engage a catch with main housing portion **14**. FIG. 2D depicts keystone assembly **30** in a raised position above main housing portion **14** having security locking device **36** accessible through a security opening **60**. Guide post **44** extends down into guide **62** to align keystone assembly **30** so that latch **46** engages with catch **64**. In the example embodiment, a releasing member **58**, such as a screw driver, inserts through security opening **60** to press against the interior member for lateral sliding movement relative to the exterior member, thereby releasing latch **46** from catch **64**. The lateral sliding motion presses ramp member **48** against a ramp surface **65** to lift keystone assembly **30** out of main housing portion **14**. Once keystone assembly **30** is removed, a disassembly of the components from within main housing portion **14** may be performed.

(36) Referring now to FIG. 3, an upper perspective view depicts main housing portion **14** when housing cover **20** lifts free after removal of the keystone assembly. A motherboard **70** couples to main housing portion **14** and interfaces a central processing unit (CPU) **72** that executes instructions to process information with a random access memory (RAM) **74** that stores the instructions and information. A pair of speakers **76** couple to main housing portion **14** and interface with CPU **72** to present audio information as audible sounds. A battery **78** is disposed between speakers **76** and interfaced with motherboard **70** provide power to the CPU, RAM and other processing components. A cooling fan **80** is disposed in main housing portion **14** to one side of motherboard **70** to generate a cooling airflow that removes excess thermal energy from the processing components. In the example embodiment, housing cover **20** includes members that extend from the housing cover bottom surface into the main housing portion interior to hold components in place. Thus, when the keystone assembly is removed to free housing cover **20**, the components under the housing cover that are held in place by members extending down from the housing cover are also released for disassembly.

(37) In the example embodiment, a speaker lock member **88** aligns with a speaker lock feature **84**, such as a cavity, notch or opening, of each speaker **76** to hold the speaker in place. A battery lock member **90** aligns with a battery lock feature **86**, such as a cavity, notch or opening, of battery **78** to hold the battery in place. In an example embodiment, battery lock member **90** may engage with a battery switch included in battery lock feature **86** that enables power transfer to and from the battery so that battery transfer is disabled when housing cover **20** is lifted off. A fan lock member **92** aligns with a fan lock feature **82**, such as a cavity, notch or opening, of cooling fan **80** to hold

the cooling fan in place. In one embodiment, the cavity, notch or opening that accepts the lock member may also offer a location for physical manipulation of the component, such as by a robotics arm. In addition to hold components in place, the locking members also prevent assembly of the housing cover **20** and components in an improper or misaligned manner since the locking members will interfere with assembly when misaligned. As described below in greater detail, a sliding engagement of housing cover **20** guides rails **96** to main housing portion **14** guides **98** aids in the assembly alignment process. In embodiments that have plural cooling fans, plural speakers and plural battery modules, a lock member may extend down aligned to hold each cooling fan, speaker and battery module in place. For instance, as is described in greater detail below, plural fans and plural battery modules may couple in series with each fan and each battery module held in place by a lock member.

(38) Referring now to FIGS. **4** and **4A** through **4C**, lower rear perspective and cross-sectional views of housing cover **20** depict coupling components within the main housing portion by members extending down from the housing cover **20**. FIG. **4** depicts a lower perspective view of housing cover **20** with a heat pipe assembly **94** coupled at the location of a CPU and proximate RAM **74**. Speaker lock member **88** inserts into speaker lock feature **84** to hold speaker **76** in place; fan lock member **92** inserts in fan lock feature **82** to hold cooling fan **80** in place; and battery lock member **90** inserts in battery lock feature **86** to hold battery **78** in position. FIG. **4A** depicts a cross-sectional view of housing cover **20** having battery lock member **90** configured to slide onto battery lock feature **86** when housing cover **20** slides into a coupled position onto main housing portion **14**, as described below in greater detail. FIG. **4B** depicts a cross-sectional view of housing cover **20** having cooling fan **80** lock feature **82** aligned to slide into speaker lock member **88** when housing cover **20** slides to couple to the main housing portion. FIG. **4C** depicts a cross-sectional view of housing cover **20** having speaker **76** speaker lock feature **84** couple against speaker lock member when housing cover **20** slides into place on the main housing portion **14**. In the example embodiment of FIG. **4C**, spring-biased contacts **100** of speaker **76** also interface with power contact pads formed in speaker lock member **88** to provide speaker signals that output audible sound from speaker **76**.

(39) Referring now to FIG. **5**, a bottom perspective view depicts housing cover **20** configured to couple in a sliding relationship with a main housing portion. Guide rails **96** on each of opposing sides of housing cover **20** insert into guides of the main housing portion and slide from a rearward to a forward position so that lock members extending from the bottom surface of housing cover **20** in the main housing portion interior engage with their associated components, such as the battery, cooling fan and speakers. When housing cover **20** slides fully forward, coupling stops **99** at the front of housing cover **20** engage with the main housing portion to hold housing cover **20** in place. Attachment of the keystone assembly at the rear side of housing cover **20** then locks the housing cover in place by preventing a rearward sliding motion that release coupling stops **99**. In the example embodiment, a keyboard membrane printed circuit board **126** extends as part of the keyboard membrane to the lower surface of housing cover **20** with a connector interface **130** aligned to couple to a motherboard connector to communicate between the keyboard membrane and the motherboard. FIGS. **6A** and **6B** depict a detail view of housing cover **20** having stops **97** that complete coupling of the housing cover **20** to main housing portion guides. Each stop **97** is positioned to engage a guide of the main housing portion as the housing cover slides forward and into position. Once engaged, stops **97** prevent lifting of the housing cover **20** so that insertion of the keystone assembly to prevent rearward sliding of housing cover **20** locks the housing as a complete assembly. Although the example embodiment depicts rails in the housing cover that engage with rail guides of the main housing portion, in an alternative embodiment, the rails may be arranged in the main housing portion to engage rail guides of the housing cover.

(40) Referring now to FIGS. **7**, **7A** and **7B**, a speaker **76** is depicted having speaker signals to play audible sounds communicated through a keyboard membrane and isolated spring biased contact

pins **100**. FIG. 7 depicts speaker **76** having spring-biased contact pins **100**, such as pogo pins, configured to align with contact pads **102** that extend down from housing cover **20** as part of the speaker lock member **88**. As shown in greater detail below, speaker contact pads **102** interface with the system motherboard through the keyboard membrane. A portion of the keyboard membrane that integrates speaker contact pads **102** insert into speaker lock member **88** to align with speaker spring-biased contact pins **100**. As is described above, speaker lock member **88** extends down from housing cover **20**, which slides from rearward to forward to couple to the main housing portion. The sliding motion also presses speaker lock member **88** against speaker **76** to engage grooves **104** formed in the speaker **76** with sidewall extensions of speaker lock member **88**. Spring-biased contact members **100** press against speaker contacts **102** to maintain an electrical interface to transfer speaker signals even with vibration associated with playing of audible sounds by speaker **76**. As is described in greater detail below, the speaker signals may be generated by an audio processor coupled to the motherboard that sends the speaker signals through the membrane. Alternatively, the audio processor may couple to a printed circuit board formed in the keyboard membrane that converts audio information provided from the motherboard into audio signals that play at speakers **76**.

(41) FIGS. 7A and 7B provide a more detailed view of the speaker signal interface between contact pads **102** of the keyboard membrane and spring-biased contact pins of speaker **76**. In the example embodiment, tapered cones **103** of dampening material, such as rubberized plastic, provide dampening of vibrations between speaker **76** and the keyboard membrane. Tapered cones **103** of dampening material isolate vibrations between the speaker and contact pads to help enhance the life of the speaker lock member, membrane contact pads and spring-biased contact pins. FIG. 7B depicts a cross-sectional view of speaker **76** depicting spring-biased contact pins **100** inserted through a speaker dampener **105** disposed around the base of spring-biased contact pins **100**. Speaker dampener **105** is formed from dampener material, such as a rubberized plastic or silicon, that absorbs vibrations to prevent wear from the vibrations on the spring-biased contact pins. In the example embodiment, some spacing is provided between speaker dampener **105** and tapered cones **103** so that each separately dampens vibrations on both sides of the electrical interface. In alternative embodiments, dampener **105** may have a cone outer shape that inserts into the cone inner shape of tapered cone **103**. The mechanical vibration dampening of the dampening material reduces wear over time related to vibrations from sound output by speaker **76**.

(42) Referring now to FIG. 8, an exploded view of a keyboard **22** depicts a keyboard membrane that includes key touch sensors, a touchpad sensor and a printed circuit board to manage logical tasks associated with keyboard **22**. In the example embodiment keyboard **22** has key assembly **112** with plural keys that couple over a planar sheet **114** having an opening for each key so that key presses are applied to a keyboard membrane **118** for detection by key input sensors at each key position, such as a switch that closes when pressed by a key. The example embodiment includes a fingerprint reader module **110** is included in one corner of the key assembly. Keyboard membrane **118** rests over housing cover **20**, which provides support so that key presses into membrane **118** will be detected by key press sensors included in the membrane. Keyboard membrane **118** has two contiguous portions formed in a common membrane film. A first portion **120** includes key touch detection sensors, such as conventional keyboard membrane switches that output a matrix location of the key press through wirelines formed in the membrane material. A second portion **122** includes a capacitive touch sensor that extends towards the front side of housing cover **20** to provide a touchpad input surface. A touchpad stiffener plate **124**, such as a hard plastic, is disposed under touchpad second portion **122** to provide a solid surface against which touchpad inputs are made. A palm rest **116** couples over second portion **122** of membrane **118** to provide a touchpad **24** that the end user can reference to make touchpad inputs over the touchpad sensor area. The key assembly **112**, planar sheet **114**, membrane **118**, palm rest **116**, stiffener plate **124** and housing cover **20** are coupled into a single keyboard assembly **125** that slides onto and off of the main housing portion

with assembly and disassembly completed by coupling or decoupling the keystone assembly to overlap keyboard assembly **125**, as described above.

(43) Referring now to FIG. **9**, an upper perspective view depicts keyboard membrane **118** having an integrated printed circuit board with a keyboard matrix controller and capacitive touch controller to manage keyboard and touchpad operations. In the example embodiment, printed circuit board **126** is a flexible printed circuit board having wirelines and contact pads formed in the membrane material. A keyboard matrix controller **127** couples to flexible printed circuit board **126** and interfaces with the matrix of keyboard touch sensors in the keyboard first portion **120** of membrane **118**. For example, keyboard matrix controller **127** decodes signals received from membrane wirelines to determine which key is pressed and codes the key value to communicate to a CPU as an input. A capacitive touch controller **121** interfaces with the capacitive touch sensor of second membrane portion **122** to report touchpad inputs to the CPU. Although the example embodiment depicts the capacitive touch controller **121** distal printed circuit board **126**, in an alternative embodiment it may couple to printed circuit board **126** or be managed with the same processing resource that detects key inputs. A communications hub **128**, such as a USB hub, interfaces with both the keyboard matrix controller and the capacitive touch controller so that the keyboard and touch inputs are communicated to the CPU with a reduced number of pins in a connector interface **130**, as is described in greater detail below. The USB protocol communicates through a differential serial pair to provide a more rapid communication transfer in the place of multiple less capable communication pathways that separately support each device or type of information. Flexible printed circuit board **126** folds under first portion **120** of membrane **118** so that connector interface **130** aligns to fit in a motherboard connector when the housing cover slides into position on the main housing portion. In various embodiments, the single piece membrane having key input sensors, capacitive touch sensors and an integrated circuit board may also include a backlight with microLEDs integrated in the membrane and controlled with a controller coupled to the circuit board, such a processing resource that coordinates keyboard operations. In the example embodiment, speaker contact pads **102** integrate with membrane **118** and fold over to fit in the speaker lock member as discussed above. An audio processor **119** couples to a printed circuit board of membrane **118** to process audio information communicated from the CPU. The result of the keyboard assembly is a set of easily assembled components that are readily reused or recycled based upon the condition of the components when the information handling system is returned.

(44) Referring now to FIGS. **10** and **10A** through **10C**, example embodiments depict keys **134** assembled as a key assembly **112** lattice with a planar sheet **114** between the keys and the keyboard membrane **118** to enhance key and keyboard reuse. FIG. **10** depicts an exploded view of planar sheet **114** above housing cover **20** having the keyboard membrane **118** coupled to an upper surface. Planar sheet **114** is a thin and flexible plastic material, such as mylar, that couples with slots **140** to connector clips **138** at opposing sides of housing cover **20** and between key assembly **112** and keyboard membrane **118**. Snap connector extensions **131** extend from housing cover **20** through slots of planar sheet **114** to engage snap connectors in the bottom surface of key assembly **112**. In the example embodiment, one snap connector extension **131** is located at each opening of planar sheet **114**, which coincides with the location of key input sensor locations. When key assembly **112** is coupled as a lattice to housing cover **20**, the keys are disassembled as a lattice by pulling up at one end of planar sheet **114** to release slot **140** from connector clips **138** so that the separation of planar sheet **114** from keyboard membrane **118** releases the lattice from snap connector extensions **131**. The proposed solution allows the keyboard assembly lattice to be readily removed and replaced at the housing cover and without impacting underlying electronics. In particular, at system refurbishment keys are typically a primary cosmetic concern. The independent lattice structure from a single material allows the lattice to remain a single component during refurbishment.

(45) FIGS. **10A**, **10B** and **10C** depict example arrangements for coupling the keyboard assembly to the housing cover with serial locking snaps that protrude through the membrane and have

substructure reinforcement metal for a robust assembly. FIGS. **10A** and **10B** depict an upper perspective view of different embodiments of snap connector extensions **131** formed in housing cover **20** that each couple to a key support or, alternatively, a lattice support. FIG. **10A** depicts a double clip arrangement that grasps a key support; FIG. **10B** depicts a single clip arrangement that couples the key support below the clips. FIG. **10C** depicts a key **134** coupled to a scissors **136** that manages vertical motion of key **134** relative to a holder plate **142** having a snap clip that couples to snap connector extension **131** extending up from housing cover **20**. Planar sheet **114** rests between membrane **118** and key **134** so that an upward lift of planar sheet **114** will release key **134** from housing cover **20**. An opening in planar sheet **114** provides to fit a rubber dome **133** through that biases up key **134** away from key input sensor **132** formed in keyboard membrane **118**. The example embodiment is described as having keys **134** coupled as a key assembly in a lattice so that lifting up on planar sheet **114** will release the keys; however, in an alternative embodiment each key **134** may couple to housing cover **20** as an individual element rather than assembled in a lattice. Generally, assembly in a lattice offers a more simple assembly and disassembly of keys **134**, however, individual elements may be supported by robotic assembly and offer advantages with respect to refurbishment of individual keys rather than recycling an entire lattice assembly when one key is bad. When recycling is necessary, the separable lattice offers efficiencies by manufacture of the entire key lattice from a single recyclable material.

(46) Referring now to FIG. **11**, an upper perspective view of a keyboard membrane **118** depicts an example embodiment having liquid detection sensors **160**. Liquid detection sensors **160** are exposed at the upper surface of keyboard membrane in the first portion **120** and interfaced with a processing resource of integrated printed circuit board **126**, such as the keyboard matrix controller. In an alternative embodiment, liquid detections sensors **160** may also be included in second portion **122** and interfaced with the capacitive touch sensor or other type of processing resource. One example liquid detection sensor **160** is a set of plural pads that detect liquid when a short is created between the pads. Another type of liquid detection sensor is a resistive sensor that has a change in resistance when a liquid comes into contact with the sensor. The contiguous nature of keyboard membrane **118** across both the key input and capacitive touch portions provides an area for liquid detection that encompasses most of the key upper surface. When liquid is detected, the liquid detection event is stored in non-transitory memory, such as flash on printed circuit board **126** or the system motherboard by communication through connector interface **130**.

(47) Referring now to FIG. **12**, a block diagram depicts liquid detection logic to detect and store a liquid spill event at a portable information handling system. In the example embodiment, liquid detection circuit **160** at detection of a liquid sends a high signal to a GPIO of keyboard controller **162**, such as a key matrix controller, which wakes in response to the GPIO input. Keyboard controller **162** writes a liquid spill event to a flash memory **164** events table **166** so that the liquid spill event remains stored in the keyboard membrane as a reference for reuse of the membrane. In addition, the event may be communicated to a non-transitory memory of the motherboard so that the liquid spill event remains with the system in case the keyboard membrane is separated and reused in a different system. As one example, events table **166** may be stored with a housing intrusion system that monitors ingress to the housing interior, as described below.

(48) Referring now to FIG. **13**, a block diagram depicts an intrusion detection system to detect and store intrusion events within the system housing. A battery management unit (BMU) **190** interfaces across a battery side interface **200** with a motherboard side interface **202** and a keyboard membrane on housing cover **20**. A battery ground through resistor **198** is located at motherboard side **202** and a sys_pre ground **192** is located at the keyboard membrane. A communications hub **194**, such as a USB hub, provides an interface between motherboard side **202** and processing resources located on the membrane printed circuit board, such as a fingerprint recognition module **196**, a second membrane portion **122** having a capacitive touch sensor to support a trackpad and a keyboard assembly **120**. Communications hub **194** provides a central communication resource with the

motherboard so that a pin count between the keyboard membrane and motherboard is reduced. Resistor **198** on the motherboard provides a PCBA level test and is do not populate (DNP) after the PCBA power test. Sys_pre ground **192** has a similar design logic on the keyboard membrane PCBA. An internal pull up pulse that is provided for power savings on sys_pre ground **192** pin in battery BMU **190** is leverage to detect a system access. Once sys_pre ground **192** pin is deasserted by the host side, such as either the motherboard or membrane, more than a threshold duration, the battery turns the power one.

(49) In the example embodiment, system intrusion is detected without a dedicated system intrusion switch or sensor. System power is cutoff when the housing cover **20** is removed so that a system intrusion event may be recorded in the BMU or with CMOS battery power at a flash of the motherboard and/or keyboard membrane circuit board. The system prevents power disruption and potential damage during assembly and disassembly and provides BMU intelligence to manage intrusion detection. Before removing power to the system, BMU **190** logs the intrusion event and can send a notification of the intrusion and system location to a remote tracking website for warranty and service tracking. In addition, communications hub **194** condenses the number of pins used to communicate between the keyboard membrane and motherboard so that a contact interface will support the communication. For instance, 60 pins associated with conventional communication of key input, touchpad, power ground and other information is reduced to 12 pins by leveraging the communication hub protocol, such as USB.

(50) Referring now to FIGS. **14** and **14A**, an example of disposition of keyboard assembly to motherboard connectors with intrusion detection are depicted. In the example embodiment of FIG. **14**, the keyboard assembly is depicted exploded up from the main housing portion **16** to show the position of the connector **204** coupled to the motherboard **70** and aligned to meet the keyboard membrane connector interface. Battery **78** couples in main housing portion **14** between speakers **76** to provide power to motherboard **70** through a battery connector as described in greater detail below. FIG. **14A** depicts a detailed view of connector **204** aligned to communicate with the keyboard assembly when coupled over the main housing portion. Communication between the keyboard membrane and motherboard is established when the cover housing slides into position over the main housing portion. BMU **190** interfaces with connector **204** through the motherboard to compare the Ra ground interface **198** and the keyboard assembly sys-pre ground **192** as described above so that an open main housing portion is detected by the BMU when the keyboard assembly lifts away from the main housing portion.

(51) Referring now to FIGS. **15** and **15A** through **15C**, an example of a 12 pin keyboard membrane connector interface with integrated intrusion detection is depicted. FIG. **15** depicts cover housing **20** lifted from main housing portion **14** to show a bottom surface having the keyboard membrane printed circuit board **126** coupled at the bottom surface with connector interface **130** exposed in a position to align with connector **204** on motherboard **70**. As described above, printed circuit board **126** is formed in the membrane material and then folded under housing cover **20** to couple to a structure on the bottom side of housing cover **20**. FIG. **15A** depicts a detailed view of printed circuit board **126** folded so that connector interface **130** couples into a defined position with twelve contact pads **212** aligned to engage against spring fingers of connector **204**. FIG. **15B** depicts connector **204** coupled to the motherboard and having twelve spring fingers **214** extending up and aligned to press against the connector interface contact pads **212**. FIG. **15C** depicts the pin positions for the spring fingers and contact pads to communicate information, power and ground between the keyboard membrane and motherboard relying in part on the USB protocol of the communications hub to communicate the information. The first four pins communicate speaker signals with positive and negative power feeds that drive the analog speaker at the speaker communication pads described above. Pin **5** provides power to the keyboard backlight, such as under the control of the information handling system embedded controller that varies the power to adjust brightness. Pin **6** is a power key switch to power the system on and off, such as power button

in the keyboard assembly. Pin **7** communicates touchpad information. Pin **8** provides a ground. Pins **9** and **10** support USB communication and can include key inputs, touchpad inputs and any other information that fits within the available bandwidth. Pin **11** supports the fingerprint reader and pin **12** supports the battery present signal described above that is also used to detect intrusion into the housing.

(52) Referring now to FIG. **16**, a circuit block diagram depicts interfaces between an information handling system motherboard and keyboard printed circuit board. An audio amplifier **222** provides speaker signals with power to play the speakers through connector **204** to the speaker connection **224**, such as the contact pads formed in the keyboard membrane and extending down the speaker lock member. In alternative embodiments, an audio processor and amplifier may couple to the keyboard printed circuit board so that audio information communicated by USB is converted to speaker signals at the keyboard membrane. A battery present ground **220** interface couples to one pin of connector **204** to detect intrusion into the housing as describe above. Communications hub **194** supports USB 2.0 to coordinate communication between the motherboard and a finger print recognition module **196** and a keyboard processing resource **228**, such as an MCU. A power regulator **230** accepts power from connector **204** to power keyboard processing resource **228**. A touchpad processing resource **226**, such as an MCU, interfaces with keyboard processing resource **228** to communicate touchpad inputs to connector **204** as USB information. Keyboard processing resource **228** interfaces with keycap LEDs **232** to backlight the keyboard, a key scan matrix **234** to detect key input values and a backlight controller **236** to manage backlight brightness. In the example embodiment, the keyboard membrane material is used to define the flexible printed circuit board that includes the processing resources on the keyboard side. In an alternative embodiment, other types of materials may be used to create the printed circuit board, which is then interfaced with the keyboard membrane.

(53) Referring now to FIG. **17**, an upper perspective exploded view of the keyboard assembly depicts disassembly of a palm rest with electrically reactive adhesive. In the example embodiment, palm rest **116** is coupled to housing cover **20** with electrically reactive adhesive **242** adjacent to key assembly **112**. Strips of tape of electrically reactive adhesive **242** are strategically placed to support movement of the touchpad **24**, such as for trackpad input clicks detect as keyboard membrane second portion **122**. Electrical contacts are interfaced with the electrically reactive adhesive so that, when disassembly is desired, a current applied to the electrical contacts will release palm rest **116**. For example, electrical contacts on the bottom side of housing cover **20** allows a current application after removal of housing cover **20** to disassemble palm rest **116**. In one embodiment, key assembly **112** may be held in place in part by an overlap of palm rest **116** to help prevent unauthorized disassembly. Alternatively, palm rest **116** and key assembly **112** may assemble and disassemble separately. In another example embodiment, system power may be made available to the contacts so that a processing component command can apply power to remove the palm rest. For example, the operating system, BIOS and/or firmware may command power to the contacts through the CPU, embedded controller and/or keyboard printed circuit board processing resource. Thus, disassembly of palm rest **116** from housing cover **20** may be limited to after removal of housing cover **20** from the information handling system or may be available while housing cover **20** remains coupled to the main housing portion. Although the example embodiment depicts strips of tape of electrically reactive adhesive **242** coupled to housing cover **20**, in various embodiments, one or both sides of the tape may have controllable adhesive. That is, a permanent adhesive may couple the strips of tape to palm rest **116** so that application of electrical current releases the palm rest from the housing cover while keeping the tape coupled to the palm rest. Alternatively, the tape may remain coupled to the housing cover while releasing the palm rest. In some cases dual sided tape may release completely from both the palm rest and housing cover when current is applied. Contacts to accept application of current to the electrically reactive adhesive may be placed in the interior of the housing so that the tape can be charged and released once the housing cover is

removed.

(54) Referring now to FIGS. **18** and **18A**, a circuit block diagram depicts keyboard assembly and disassembly of a palm rest with electrically reactive adhesive. In the example embodiment, an embedded controller **262** on a system motherboard and executing firmware in cooperation with the system BIOS applies current from the system power supply **264** to tape contacts **266** to release the electrically reactive adhesive **242** tape. In an alternative embodiment, embedded controller **262** coordinates release of the tape by applying current to tape contacts **266** that are formed as part of the keyboard membrane, such as with a command communicated to the keyboard controller of the keyboard printed circuit board. FIG. **18A** depicts an example of electrically reactive adhesive tape that is released by application of current. An anodic metal substrate **246** is separated from a cathodic conductive substrate **250** by an adhesive with cationic additive **248** having cations in ionic liquid as depicted. Application of an electrical stimulus **260** results in electrochemical reactions at the interfaces that trigger interfacial debonding with movement of cations away from the anode breaks the metal-adhesive bond. As is described above, the adhesive may have one or two sides. In one embodiment, a two-sided adhesive may separate electrical interfaces so that each side is released with a separate application of current. Thus, for example, one side of tape is released to release the palm rest from the housing cover, then the second side of tape is released to release the tape completely from the surface. An end user can select whether to release the palm rest from the tape first or the housing cover.

(55) Referring now to FIGS. **19A** through **19C**, a cooling fan **80** coupling with a motherboard **70** in a main housing portion **14** secures the motherboard in place. Motherboard **70** couples to main housing portion **14** with retention nuts **320** that extend up from main housing portion **14** to fit through openings in motherboard **70**. The openings have a pear shape so that when the motherboard slides towards the rear of the housing a lip of the retention nut retains the motherboard in position, as described in greater detail below. Battery **78** fits onto a battery connector to provide power to the motherboard, as detailed below, and speakers **76** rest in a speaker cavity aligned to interface with the speaker contact pads that extend down when the housing cover couples over main housing portion **14**. Before cooling fan **80** slides into position as indicated by arrow **270**, each of motherboard **70**, speakers **76** and battery **78** are free to be lifted from main housing portion **14**, such as with a robotic arm. Cooling fan **80** slides as indicated by arrow **270** so that a cooling fan connector **272** fits under motherboard **70** to couple to a motherboard cooling fan connector **280**, which provides power and fan speed control to cooling fan **80**. Cooling fan **80** engages in structures of main housing portion **14**, such as against vents **282**, so that engagement of cooling fan **80** with motherboard **70** prevents forward sliding of motherboard **70**, thereby securing the motherboard to main housing portion **14** with engagement of the lip of retention nuts **320**.

(56) FIG. **19B** depicts cooling fan **80** fully slid to the side of motherboard **70** so that connectors **272** and **280** fully engage to establish operational power and control communication between motherboard **70** and cooling fan **80**. A rib structure **276** of cooling fan **80** inserts into a slot **278** of motherboard **70** to ensure proper alignment and reinforce against sliding motion of motherboard **70**. Once cooling fan **80** fully slides into position, airflow is drawn by the cooling fan and directed out exhaust vent **282** to provide thermal rejection for motherboard processing components, such as the CPU. FIG. **19C** depicts the alignment of rib structures **276** and slots **278** to provide a robust structure against sliding movement of motherboard **70**. A clip **274** formed in cooling fan **80** proximate the connector **272** engages with motherboard **70** to lock the cooling fan in a relative vertical position. At disassembly, pressing on clip **274** and pulling cooling fan **80** to slide away from motherboard **70** releases the motherboard for removal from the main housing portion **14**, such as with manipulation with a robotic arm to slide and release from retention nuts **320**.

(57) Referring now to FIGS. **20** and **20A** through **20E**, an information handling system having a series cooling fan interconnect with adjustable cooling capacity is depicted. FIG. **20** depicts an exploded view of first and second cooling fans **80** aligned to couple to main housing portion **14** to

support cooling airflow for processing components coupled to a motherboard **70**, such as a CPU. A first cooling fan **80** interfaces a connector **272** of the cooling fan to a connector **280** at the bottom surface of motherboard **70** as described above. An exhaust **282** includes a cooling channel **283** that directs cooling airflow pulled across motherboard **70** through cooling channel **283** to exchange thermal energy with a heat sink and out exhaust **282**. A second cooling fan **80** has a connector **272** that couples with a rear connector **280** on the bottom side of the first cooling fan and having the same footprint as the motherboard connector. The second cooling fan in the example embodiment exhausts airflow out exhaust **282** without a cooling channel **283**. Cooling fans **80** under the management of an embedded controller on motherboard **70** manage fan speeds relative to each other to optimize how cooling air flows through the housing and out of exhaust **282**. For example, the embedded controller of motherboard **70** detects the presence of multiple cooling fans and sends control commands through a series connection to each cooling fan. FIG. **20A** depicts the first and second cooling fans assembled in series with the first cooling fan blowing cooling airflow across cooling channel **283** and the second cooling fan blowing cooling airflow across a gap and out of exhaust **282**. The cooling fan speeds may vary based upon sensed exhaust temperatures and other internal temperatures. For instance, a moderate CPU temperature could tolerate lower airflow through cooling channel **283** to achieve higher overall airflow with the lower resistance to air exhaust of the second cooling fan. Although the example embodiment depicts two cooling fans, in alternative embodiments any number of cooling fans may be interfaced in series provided room exists in the housing.

(58) FIG. **20B** depicts a side view of motherboard **70** having a connector **280** on the bottom side and aligned to couple with the connector **272** of cooling fan **80**. A second fan **80** interfaces serially with motherboard **70** through the first fan by coupling a connector **272** at a front upper surface with a connector **280** on the first fan bottom rear surface that has the form factor of the motherboard connector **280**. FIG. **20C** depicts a side view of the two cooling fans **80** serially coupled to each other so that a processing component of the motherboard, such as an embedded controller, manages both cooling fans by communicating with the farthest fan through the intermediary fan. The outer most cooling fan draws cooling airflow through the intermediary cooling fan to increase the cooling airflow across motherboard **70** or, alternatively, draws air from the housing interior to help decrease interior air temperatures. FIG. **20D** depicts a perspective transparent view of the first cooling fan **80** connector **272** having spring fingers that bias upward against contact pads of the motherboard **70** connector **280**. FIG. **20E** depicts a perspective transparent view of the second cooling fan **80** having a connector **272** with spring fingers biased up against connector **280** contact pads formed in the bottom surface of the middle fan **80**. In alternative embodiments different numbers of cooling fans with varying capabilities and air flow configurations may be used in series to achieve a desired thermal and acoustic profile for the information handling system.

(59) Referring now to FIG. **21**, a circuit block diagram depicts an example of communication paths between serially interfaced cooling fans. In the example embodiment, a common power and ground pin at each cooling fan **80** shares power and ground provided from the motherboard. Pins **2** and **3** provide pulse width modulation (PWM) control and tachometer feedback for the first cooling fan. Pins **4** and **5** provide PWM control and tachometer feedback for the second cooling fan. Each of the pins interface with an embedded controller of the motherboard that executes thermal management firmware to command a fan speed for each cooling fan. In one embodiment, additional cooling fans are supported by adding a PWM and tachometer pin for each additional cooling fan. In an alternative embodiment, a SPI, UART or I2C interface may be used in the place of the PWM interface to control multiple cooling fans.

(60) Referring now to FIGS. **22**, **22A** and **22B**, an example embodiment depicts a cable-free battery connector to communicate power between an information handling system motherboard and battery. FIG. **22** depicts the information handling system with motherboard **70**, cooling fan **80** and battery **78** exploded out of main housing portion **14**. A battery connector **300** is coupled to main

housing portion **14** between the coupling locations of battery **78** and motherboard **70**. Battery **78** is placed in main housing portion **14** aligned to have contact pads on the battery bottom surface interface with spring members **302** on one side of battery connector **300**. Once battery **78** rests in place on main housing portion **14**, a lock member extending down from the battery cover couples the battery in place. Motherboard **70** has openings as described above that couple to attachment members extending up from main housing portion **14** by sliding motherboard **70** forward to engage on lips of the attachment members. When motherboard **70** slides into the engaged position, contact pads **301** on the bottom surface of motherboard **70** interface with spring members **302** of battery connector **300** to communicate power with the battery. The position of battery connector **300** and spacing between the spring members provides room to slide motherboard **70** forward when battery **78** is in place. Similarly, when motherboard **70** is engaged with the attachment members, battery **78** removes from main housing portion **14** by lifting out without impacting motherboard **70**. In this manner, automated assembly and disassembly is supported, such as by robotic arms. FIG. **22A** depicts a side view of motherboard **70** bottom contact pads **301** and battery **78** bottom contact pads **301** align to couple with spring clips of battery connector **300** when assembled in main housing portion **14**.

(61) FIG. **22B** depicts an example of battery **78** that provides programmable parallel/serial battery back extensions to reuse, recycling and performance level adjustments. In the example embodiment, battery **78** is expandable by first and second battery packs **294** that fit in main housing portion **14** and electrically couple through opposing connector spring clips **292** and connector pads **290**. In the example embodiment, each battery pack **294** has an arm that extends spring clips **292** towards a parent battery interface so that the outer side of the battery pack fits within the main housing portion. Spring clips **292** and contact pads **290** extend normal the battery and motherboard interface, although an alternative embodiment may have a linear arrangement based upon the type of shape of the main housing portion and position of the motherboard. The electrical interface and battery configuration may depend based upon the type of system and the system operating conditions. For instance, battery **78** BMU or the motherboard embedded controller may arrange the batteries to have a parallel interface with a reduced voltage and deeper current generation or a series interface with an increased voltage and lesser current generation. The voltage output of battery **78** may vary by changing one or more of the battery packs between series and parallel configurations when operating conditions that vary at the information handling system, such as during surges of power use due to heavy processor use or charging of the battery with external power available. Communication between the motherboard, the battery and the battery pack is managed with a BMU at each battery and communication of commanded battery configurations from the motherboard through each battery and battery pack in series. In one embodiment, CPU performance is managed based upon battery configuration for available voltages and currents, such as by throttling and surging the CPU as desired with corresponding adjustments to battery configurations. Similarly, configuration between series and parallel may help to extend the life of a battery and battery pack arrangement when a battery cell is detected as bad. Thus, for instance, a BMU may command an embedded controller to adjust CPU operating conditions based upon the battery configuration.

(62) Referring now to FIG. **23**, an exploded front perspective view depicts the cable-free battery connector. A bottom portion **306** couples to the main housing portion and accepts metal spring fingers **302** that conduct power, ground and BMC-to-EC communications. A top portion **304** couples to bottom portion **306** to capture metal spring fingers **302** in place. Alignment posts **308** extend up from bottom portion **306** and above top portion **304** to insert in the battery and motherboard to ensure proper interfaces for the metal spring fingers. Alignment structures **310** extend out of both sides of bottom portion **306** as an alignment reference for mounting to the main housing portion. Top portion **304** clips into place on bottom portion **306** with resilient snaps **312** that fit into and engage with slots **314** so that connector **300** will not disassemble when coupled in

an information handling system since the release is hidden from the end user. When an information handling system is refurbished, connector **300** may be reused in an assembled manner, broken down to replace or recoat metal spring fingers **302**, or broken down to recycle the plastic and metal as separate materials with an automated break down process, such as interaction with robotic tools. (63) Referring now to FIG. **24**, a front perspective view depicts battery connector **300** coupled to the main housing portion and having motherboard **70** coupled in place to electrically interface with the battery. Alignment post **308** fits through a slot of motherboard **70** so that motherboard **70** can slide over top of connector **300**. Retention nut **320** fits into the wide portion of pear-shaped slot **322** so that motherboard **70** engages under a lip of retention nut **320** when slid forward. As is described above, motherboard **70** is then held in position to prevent rearward sliding by engagement of a cooling fan at one side of the motherboard. Keyboard connector is carried with motherboard **70** to align with the keyboard once the cover housing is coupled into place, thereby completing the battery connect signal. Connector **300** couples in an aligned position with alignment structure **310** fit into a coupling structure of main housing portion **14**. The fit and alignment structures support fully automated assembly and disassembly of the information handling system, such as with robotic arms. Further, the components are built in a modular manner that encourages reuse of those having longer useful life and recycling of like materials after system breakdown.

(64) Referring now to FIGS. **25**, **25A** and **25B**, an example of assembly of components with retention nuts **320** is depicted. FIG. **25** depicts an example of distribution of retention nuts **320** on the bottom surface of main housing portion **14** as desired to secure components to main housing portion **14**. To aid automated assembly of information handling system **10**, retention nuts **308** couple in position with a press fit or, alternatively an eye hole type of retention. FIG. **25A** depicts a side view of retention nut **320** that couples in position with a clinch nut **334** having a directional element **332** at a base main body **330**. A nipple portion **324** at the top side of main body **330** has an engagement portion **326** that provides lip to fit over the motherboard or other component that is to be coupled in place. Engagement portion **326** has an overtravel feature **328** on one side that stops a component from sliding relative to retention nut **320** past a certain distance. For instance, when nipple portion **324** fits into the larger circumference portion of a motherboard pear-shaped slot, engagement portion **326** extends over the top surface of the motherboard to hold it in position with sliding movement halted by overtravel feature **328**. FIG. **25B** depicts a bottom front perspective view of retention nut **320** to illustrate clinch nut **334**, which has a desired alignment in the housing enforced by directional element **332**.

(65) Referring now to FIG. **26**, an antenna module **338** integrated in modular thermal solution is depicted. Exhaust **282** is depicted exploded from cooling channel **283** and motherboard **70**. A radio integrated circuit **334** couples to motherboard **70** and interfaces with the CPU to communicate information as wireless signals sent external to the information handling system. For example, radio integrated circuit **334** communicates with wireless local area networks (WLAN) in the 5 and 2.4 GHz bands, with wireless personal area networks (WPAN) like BLUETOOTH in the 2.4 GHz band and/or wireless wide area networks (WWAN) through commercial cell network frequency bands. A coaxial cable **332** carries the wireless signals between the radio and the antenna modules **338**. Antenna modules **338** and coaxial cables **332** integrate into a single module assembly with exhaust **282** so that antenna modules **338** maintain a desired position for transmitting and receiving wireless signals, such as embedded within a nonconductive material **336** and relative to a ground plane provided by metal of cooling channel **283**. The exhaust vent **282** as a completed module helps to optimize radio performance while leveraging the thermal solution as part of the antenna footprint for a more efficient architecture.

(66) Referring now to FIGS. **27A**, **27B** and **27C**, an alternative embodiment is depicted for coupling a motherboard to the information handling system main housing portion. FIG. **27A** depicts motherboard **70** in position to fit on main housing portion **14** over a bi-stable flexure plate **350** configured to couple the motherboard in place. Bi-stable flexure plate **350** is slid to the right

relative to a fixed position of a central portion 354 having a pin 357 or other coupling device to fix its position. Motherboard 70 is placed downward on bi-stable flexure plate 350 with a hook 352 slid to the right to provide space for motherboard 70 to rest in place, as indicated by arrow 358. FIG. 27B illustrates that once motherboard 70 rests on bi-stable flexure plate 350, a push to the right as indicated by arrow 360 overcomes the bias of the fixed central portion 354 so that hook 352 engages over top of motherboard 70. FIG. 27C depicts hook 352 coupled over motherboard 70 so that motherboard 70 is coupled in place with a bias of central portion 354 holding bi-stable flexure plate 350 in place.

(67) Referring now to FIGS. 28A and 28B, an example is depicted of a flexible slide motion arrangement for coupling motherboard 70 to the main housing portion. FIG. 28A depicts the central portion 354 in an unlocked position having a gap 362 to the right side of bi-stable flexure plate 350. Eight triangular shaped openings are formed by a structure 356 on opposing sides of central portion 354. FIG. 28B depicts that a gap 362 moves from the right side to the left side of central portion 354 when bi-stable flexure plate 350 is slid to the right as indicated by the arrows. In the example embodiment, gap 362 is approximately 4 mm. When gap 362 is on a side of central portion 354, the length of the central member of structure 356 works against movement of the gap to the other side central portion 354. In order to shift bi-stable flexure plate 350 to the locked position, the central arm between the triangle structures has to flex so that the resilient nature of the material biases the bi-stable flexure plate 350 to assume a position to the right or to the left of central portion 354 and need a substantial force to change the position. The amount of force needed to couple and release motherboard 70 is adjusted by adjusting the type and amount of material used to define structure 356, including the size of the triangles.

(68) Referring now to FIG. 29, a bi-stable flexure plate is depicted for coupling a motherboard to a main housing portion. In the example embodiment, a central pin 366 fits through an opening 370 of motherboard 70 to hold motherboard 70 in position relative to central portion 354, which is itself coupled to a fixed position relative to the main housing portion. A sliding pin 368 engages in a slot opening 372 to allow the bi-stable flexure plate 350 to slide relative to mother board 70 during locking and unlocking. Side coupling clips 364 align with slots 374 of motherboard 70 and hook 352 aligns with slot 376. A push left or right on the bi-stable flexure plate 350 changes the bias of triangle structure 356 to adjust between locked and unlocked positions relative to motherboard 70. In one embodiment, bi-stable flexure plate 350 is made of a resilient plastic by an injection mold process and pins 366 and 368 elevate mother board 70 in a raised position to aid with thermal management.

(69) Although the present invention has been described in detail, it should be understood that various changes, substitutions and alterations can be made hereto without departing from the spirit and scope of the invention as defined by the appended claims.

Claims

1. An information handling system comprising: a housing having first and second housing portions rotationally coupled by a hinge, the first housing portion having first and second rails extending into an interior of the first housing portion from opposite sides of the first housing portion; a processor disposed in the first housing portion and operable to execute instructions that process information; a memory disposed in the first housing portion and interfaced with the processor, the memory storing the instructions and information; a keyboard assembly having plural keys at an upper surface configured to accept end user key press inputs and having first and second guide structures at opposite sides aligned to engage with the first and second rails in a sliding engagement that couples the keyboard assembly to the first housing portion by sliding from a rear initial position proximate the hinge to a forward position that leaves a gap between the keyboard assembly and the hinge; a stop defined in each of the first and second guide structures, the

keyboard assembly sliding the guide structures on the rails until the stop to align the keyboard assembly is in position to enclose the first housing portion; a keystone assembly having plural latches aligned to couple with plural catches of the first housing portion proximate the hinge to fill the gap between the keyboard assembly and the hinge, the stop aligning the keyboard assembly to have an open space to the first housing portion proximate the hinges, the keystone assembly coupling in the open space and over a portion of the keyboard assembly to retain the keyboard assembly in the position over the first housing portion, the keystone assembly having an exterior member and an interior member engaged to slide within the exterior member between a first position engaging the latches and the catches and a second position releasing the latches and catches; a spring element integrated in the interior member; and a keystone stop integrated in the exterior member, the exterior member coupling in a fixed position relative to the first housing portion, the interior member sliding relative to the first housing portion when pressed, the spring element working against the keystone stop to bias the interior member to a latched position having the latches engaged with the catches.

2. The information handling system of claim 1 further comprising: a security device coupled in the first housing portion at one end of the keystone assembly and accessible through a security opening of the first housing portion; and wherein the interior member is accessible to a push to a release position through the security opening.

3. The information handling system of claim 2 further comprising: a component lock member extending from the keyboard assembly into the first housing portion interior; and a component disposed in the first housing portion interior and aligned with the component lock member, the component lock member engaging the component to retain a position of the component when the first and second rails engage the first and second guide structures to the stop.

4. The information handling system of claim 1 further comprising: a membrane disposed below the plural keys to detect key touches; and a power interface through the membrane and to the component lock member, the power interface providing power to the component.

5. The information handling system of claim 4 wherein the component comprises a speaker configured to play audible sounds based upon signals communicated through the power interface.

6. A method for assembly of an information handling system, the method comprising: assembling a keyboard to an upper surface of a housing cover; inserting a guide structure of a bottom surface of the housing cover onto a rail of the housing at a rear side of the housing; sliding the housing cover forward relative to the housing to couple the housing cover over the housing interior with engagement of the rail and guide structure with a gap between the rear side of the housing and the housing cover, the housing cover sliding until the rail hits a stop of the guide structure; coupling a keystone assembly in the gap between one side of the housing and the housing cover, the keystone assembly filling the gap so that the housing cover and keystone assembly enclose the housing interior; extending a component lock member from the bottom surface of the housing into the housing interior; in response to the sliding the housing cover, aligning the component lock member with a component disposed in the housing interior, the component lock member retaining the component in position in the housing; wherein: the component comprises a speaker having a power connector; and the aligning further comprises sliding a member extending from the housing cover bottom surface to the power connector to transfer power from a membrane of the keyboard to the speaker.

7. The method of claim 6 further comprising: engaging a latch of the keystone assembly to a catch of the housing to couple the keystone assembly to the housing; sliding an interior member of the keystone assembly relative to an exterior member, the interior member releasing the latch from the catch; and engaging by the sliding a ramp member of the interior member and the housing to bias the keystone assembly away from the housing.

8. The method of claim 6 wherein: the component comprises a cooling fan adjacent a motherboard; and the aligning further comprises sliding a member to engage against the cooling fan, the member

retaining the cooling fan in position, the cooling fan retaining the motherboard in position.

9. The method of claim 6 wherein the component comprises a battery.

10. A keyboard assembly comprising: plural keys exposed at an upper surface to accept key inputs; a housing cover supporting the plural keys at an upper surface; first and second guide structures at opposite sides of the housing cover and aligned to engage with first and second rails of a rear side of a housing in a sliding engagement that couples the keyboard assembly to the housing when the cover slides to a front side of the housing; a stop disposed in each guide structure to define the amount of sliding engagement to provide a predetermined gap at one side of the housing cover and housing; a keystone assembly configured to couple to the housing cover upper surface and the housing in the predetermined gap to retain the guide structure in place when engaged with the rail to the stop; and plural component lock member extending from the housing cover bottom surface, each component lock member aligned to engage with a component disposed in the housing to retain the component in position.

11. The keyboard assembly of claim 10 further comprising: a membrane disposed below the plural keys to detect key presses; and a power interface from the membrane to at least one of the plural component lock members to communicate power to a component associated with the at least one of the plural component lock members.
